

Nonlinear Re-Randomization and CG Diversification

Why the new gadgetize reverted to the nonlinear RGs, and why each CG is now drawn from a random menu

local_mixing project

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Abstract

The 2026-07-20b revision of the new-gadgetize recipe makes two changes to the gadget body: the linear CNOT re-randomization gadgets $\{RG2, RG3\}$ are replaced by the legacy *nonlinear* g57 networks $\{RG1, RG2, RG3\}$ (one uniform draw per shared gate), and the fixed four-fragment realization of the shared computation gadget (CG) is replaced by a per-gate uniform draw over *seven* structurally different realizations. This note records the motivation: what the degree-bounded reconstruction experiments showed about the linear design, what the reversal buys and what it deliberately gives up, and why implementation diversity of the CG is itself a security measure. Companion changes (the final commuting-order rerandomization and the sandwich column float) are summarized at the end.

1 Setting

The two-share gadget maps a circuit C on n wires to an equivalent circuit on $2n$ wires. Every logical value v is carried as an XOR pair $v = s_v \oplus p_v$; the construction consists of random Z bookends on the aux half, a W_i encode/decode curtain of CNOTs that installs and removes the sharing, one *computation gadget* (CG) per source gate, and a stream of *re-randomization gadgets* (RGs) that churn the sharing between CGs. The zero-slice CNOT/CCNOT preblock guards the input port.

The original new-gadgetize design chose *linear* RGs (a 3-CNOT re-pairing RG2 and a 2-CNOT mask-refresh RG3) and one fixed four-fragment CG. Both choices optimized a specific adversary model: *gate-local non-completeness* — no single physical gate ever reads both carriers of one value, so every prefix of the circuit is first-order probe-masked — plus gate-count leanness.

2 The adversary that changed the decision

The project's security target is indistinguishability against *all feasible distinguishers*, evaluated against concrete cheap attack families. The family that drove this revision is *degree-bounded algebraic reconstruction* (`hmap_affine`): for prefixes C_i of the original and G_j of the gadgetized circuit, fit each bit of $C_i(x)$ as a degree- d GF(2) function of *all* of G_j 's wires (train/holdout split; inconsistency scores 0.5). Low error means the original computation's progress is legible at that point of the gadget; the ideal map is flat at 0.5 except the two forced I/O corners. The

measure is invariant to affine re-encodings, which is exactly what makes it the right lens for a sharing-based gadget: *the XOR sharing itself is free for this adversary*.

Two facts came out of the measurement campaign:

1. **Degree-2 reconstruction is more persistent than degree-1 under mixing.** Mixing erodes the degree-1 (affine) diagonal quickly, but the degree-2 diagonal survives much longer. The affine conjugation twists of the mixing stage cannot address this: an affine change of variables is transparent at every degree.
2. **The gadget encodings differ sharply.** The old (legacy g57) gadget’s output is *not* recoverable even by restricted degree-2 predictors, while the linear-RG CNOT gadget is affine-transparent at its interface. Computing the ANF degree of each component isolated the cause: the sharing setup (M, W_i) is affine everywhere; the nonlinearity of the legacy gadget lives in its gate application (degree 2, once per source gate, compounding) and its re-randomizers — RG1 (degree 3), RG2 (degree 2), RG3 (degree 2). The linear replacement kept the setup and *removed exactly the nonlinear part*.

The structural point is simple: affine maps compose to affine maps. With linear RGs, the entire re-randomization stream between any two CGs is one affine transformation of the wire state, so a degree- d predictor sees through an *arbitrarily long* RG stream at no cost in degree. The stream churned the sharing but hid nothing from the adversary that matters. Nonlinear RGs break this composition: every RG multiplies algebraic degree into the transition, and reconstruction across a stretch of the body must now pay degree for every RG it crosses.

3 The reversal: nonlinear $\{RG1, RG2, RG3\}$, one per CG

The revision reinstates the legacy g57 networks, drawn uniformly at one RG per CG gap (`--rg-frequency`, default 1 on this path):

RG	effect on the sharing	gates	ANF degree
RG1	swap the virtual values of two pairs	6 g57	3
RG2	re-pair two pairs crosswise	6 g57	2
RG3	XOR $r_1 \vee \neg r_2$ of two foreign wires into both carriers	2 g57	2

What is bought. Nonlinear transitions of degree 2–3 that compound along the body; a re-randomization stream that low-degree reconstruction cannot cross for free; and, as a side effect, nonlinear material (the RG3 reads of two *random* foreign wires) distributed through the entire body rather than confined to the bookends.

What is paid, deliberately. RG1 and RG2 gates read both carriers of a value, so gate-local non-completeness — first-order probe masking — is given up. This is a conscious ranking of adversaries: the probing model was a secondary concern, while degree-bounded reconstruction is a demonstrated, cheap, working attack on the linear design. (RG3 keeps single-carrier reads; the masking-safe CG variants below keep the option of a partially masked body alive.)

Measured effect ($n=16$, pre-mixing, old vs. revised recipe): the maps flatten (degree-1 σ $0.125 \rightarrow 0.106$, degree-2 $0.161 \rightarrow 0.149$; the interior column-mean dip that leaked the temporal layout largely evens out), and the best-case interior degree-1 reconstruction error roughly doubles

($\approx 0.05\text{--}0.15 \rightarrow 0.2\text{--}0.3$). One honest caveat: the degree-2 *snapshot* diagonal persists in both versions. This is inherent to the linear XOR sharing — at any prefix the logical value *is* $s \oplus p$, a degree-1 function of the wires — independent of how nonlinear the transitions are. Erasing snapshots is the mixing stage’s job (structural DB/split mixing is what erodes it), or, in a future design, a nonlinear share *encoding*.

4 CG diversification: a seven-variant menu

The CG applies $A \oplus = B \vee \neg C$ over the shares, i.e. $f = 1 \oplus C \oplus BC$ with $B = b_0 \oplus b_1$, $C = c_0 \oplus c_1$. The original design emitted one fixed shape — four mixed-polarity conjunction fragments — for every source gate.

Why fixed is bad. A fixed CG is a syntactic beacon: a fixed period, a fixed local gate-type pattern, and a fragment census that counts the source gates exactly. Regularity is precisely what static profiling and SAT-style attacks feed on — a solver that learns one CG shape has learned them all. And the fragments themselves are cold material for the mixing stage: the frozen replacement store matches g57 windows at $\sim 99\%$ but mpmct fragment windows at $\sim 2.8\%$, so a fragment-heavy body resists exactly the phase-A mixing that is supposed to dissolve it.

The menu. Each source g57 is now emitted as one of seven realizations, drawn uniformly per gate, with the carrier roles (which carrier of A is targeted; share/pad order of B and of C) also randomized per draw:

#	name	shape	idea
0	collapse both	4 CNOT + 1 g57	$b_0 \oplus = b_1$, $c_0 \oplus = c_1$ put B , C on single wires; one g57 fires f ; restore
1	collapse c	2 CNOT + 2 g57 + $\neg$ CNOT	pair sums to BC ; $a \oplus = \neg c_0$ adds $1 \oplus C$
2	collapse b	2 CNOT + 1 g57 + 1 frag	$[a, B, c_0]$ plus fragment $\neg B \wedge c_1$
3	linear tail	$\neg$ CNOT + CNOT + 4 g57	no collapse; quad $[a, b_i, c_j]$ sums to BC ; masking-safe
4	legacy GADGET	6 g57	the classic nonlinear network
5	quad + pair	5 g57 + 1 frag	quad BC , then $(\neg c_0 \wedge c_1) + [a, c_1, c_0] = 1 \oplus C$
6	4-cube ESOP	4 fragments	the previous fixed CG; masking-safe

Vocabulary is restricted to g57s and pure conjunctions with one or two controls of any polarity. In particular, *no bare X gates*: several variants algebraically need a “ $\oplus 1$ ”, but an X census would count the source gates, so every constant is folded into a negative-control CNOT or a complemented fragment.

Gate-by-gate. Each realization writes only the target carrier a^* (one of A ’s two carriers, chosen per draw); b_0, b_1 and c_0, c_1 are the two carriers of B and of C in a per-draw random order. Notation, all over GF(2) (so $+$ is $\oplus$ and $\bar{w} = 1 \oplus w$): $t += w$ is a CNOT ($t \leftarrow t \oplus w$); $t += \bar{w}$ a negative-control CNOT; $t += (x \vee \bar{y})$ a g57; and $t += (\ell \ell')$ a two-control conjunction fragment (juxtaposition = AND, each literal w or $\bar{w}$). Recall $f = 1 \oplus C \oplus BC = B \vee \bar{C}$. Gates execute top to bottom.

0 — collapse both

(4 CNOT + 1 g57):

- | | |
|---------------------------------------|---------------------------------|
| 1. $b_0 += b_1$ | <i>now $b_0 = B$</i> |
| 2. $c_0 += c_1$ | <i>now $c_0 = C$</i> |
| 3. $a^* += (b_0 \vee \overline{c_0})$ | $= B \vee \overline{C} = f$ |
| 4. $c_0 += c_1$ | <i>restore c_0</i> |
| 5. $b_0 += b_1$ | <i>restore b_0</i> |

1 — collapse c

(2 CNOT-type + 2 g57):

- | | |
|---------------------------------------|--|
| 1. $c_0 += c_1$ | <i>now $c_0 = C$</i> |
| 2. $a^* += (b_0 \vee \overline{c_0})$ | |
| 3. $a^* += (b_1 \vee \overline{c_0})$ | <i>the two g57s sum to BC</i> |
| 4. $a^* += \overline{c_0}$ | <i>adds $\overline{C} = 1 \oplus C$</i> |
| 5. $c_0 += c_1$ | <i>restore c_0</i> |

2 — collapse b

(2 CNOT + 1 g57 + 1 fragment):

- | | |
|---------------------------------------|---|
| 1. $b_0 += b_1$ | <i>now $b_0 = B$</i> |
| 2. $a^* += (b_0 \vee \overline{c_0})$ | $= 1 \oplus c_0 \oplus Bc_0$ |
| 3. $a^* += (\overline{b_0} c_1)$ | $= c_1 \oplus Bc_1$; <i>total = f</i> |
| 4. $b_0 += b_1$ | <i>restore b_0</i> |

3 — linear tail

($\overline{\text{CNOT}}$ + CNOT + 4 g57; masking-safe, no carrier collapsed):

- | | |
|---------------------------------------|---|
| 1. $a^* += \overline{c_0}$ | |
| 2. $a^* += c_1$ | <i>gates 1-2 give $\overline{c_0} \oplus c_1 = 1 \oplus C$</i> |
| 3. $a^* += (b_0 \vee \overline{c_0})$ | |
| 4. $a^* += (b_1 \vee \overline{c_0})$ | |
| 5. $a^* += (b_0 \vee \overline{c_1})$ | |
| 6. $a^* += (b_1 \vee \overline{c_1})$ | <i>the four g57s sum to BC</i> |

4 — legacy GADGET

(6 g57; c_1 borrowed as restored scratch):

1. $c_1 += (b_0 \vee \overline{b_1})$ *scratch on c_1*
2. $a^* += (c_1 \vee \overline{b_1})$
3. $a^* += (b_0 \vee \overline{c_1})$
4. $c_1 += (b_0 \vee \overline{b_1})$ *restore c_1*
5. $a^* += (b_1 \vee \overline{c_0})$
6. $a^* += (c_0 \vee \overline{b_0})$

5 — quad + pair

(5 g57 + 1 fragment; no carrier collapsed):

1. $a^* += (b_0 \vee \overline{c_0})$
2. $a^* += (b_1 \vee \overline{c_0})$
3. $a^* += (b_0 \vee \overline{c_1})$
4. $a^* += (b_1 \vee \overline{c_1})$ *the four g57s sum to BC*
5. $a^* += (\overline{c_0} c_1)$
6. $a^* += (c_1 \vee \overline{c_0})$ *gates 5–6 sum to $1 \oplus C$*

6 — 4-cube ESOP

(4 fragments; the previous fixed CG, masking-safe):

1. $a^* += (\overline{c_1} b_0)$
2. $a^* += (c_1 \overline{b_1})$
3. $a^* += (c_0 b_1)$
4. $a^* += (\overline{c_0} \overline{b_0})$ *the four cubes XOR to $B \vee \overline{C}$*

Every non-target carrier that a variant touches (the collapses in 0–2, the scratch in 4) is written an even number of times and so is restored exactly at the wire level; verified exhaustively over all 7 variants, 8 carrier role assignments, and 64 inputs.

What diversification buys.

- *Irregularity*: no fixed CG period, no repeated shape, varying local gate-type statistics — less structure for syntactic profiling and for solvers to latch onto. The carrier-role randomization scrambles pin patterns even between same-variant draws.
- *DB-warmth*: the body census flips from fragment-heavy to g57-dominant (at $n=16$, g57 / CNOT / fragment counts $822/277/707 \rightarrow 1269/486/264$), with CNOTs only sprinkled — and a few CNOTs within many g57s are well tolerated by the replacement store, while the g57 mass mixes at the $\sim 99\%$ match rate.
- *More nonlinearity in the body*: five of seven variants carry g57 material into the CG itself, compounding with the nonlinear RG stream.

Cost: the menu averages ~ 5.1 gates per CG versus 4, i.e. $\sim 12\%$ body growth at $n=16$ — accepted.

Every variant is verified exhaustively (all 7 variants $\times$ 8 role draws $\times$ 64 carrier assignments: exact update, wire-level restoration of every non-target carrier, vocabulary compliance), and the driver’s pinned zero-slice functionality check runs on every build.

5 Companion changes, briefly

Commuting-order rerandomization (`commuting_shuffle`, 5689c8ea). The construction-time layout $S_1 | Z | W | \text{body} | W^{-1} | Z$ was legible in the heatmaps as an S-shape — no mixing between the computation block and the bookends, with the affine W_i curtain sitting between them. The emitted order is now a fresh random order that preserves only the relative order of gate pairs that *provably* collide (`XGate::collides`: “commute unless proven otherwise”, including the opposite-polarity shared-control exemption). This is a one-time, structure-dependent reorder — consistent with the principle that local mixing belongs to the mixing stage, whose job the gadget stage only sets up.

Sandwich column float (9323272a). In the sliced sandwich, each middle-column CNOT is assigned a registered random direction and floats that way to its commutation extreme, dissolving the $C|N|D$ seam — the sandwich’s most structure-revealing feature — into a band before gadgetization.

6 Open items

The production $n=128$ A/B (old vs. revised recipe through the two-phase mixing pipeline, evaluated with `hmap_affine` at degrees 1 and 2) has not yet been run. The persistent degree-2 snapshot legibility of the linear XOR sharing remains the deepest open design question; the candidate answers are a nonlinear share encoding (snapshots stop being degree-1 in the wires) and generalized nonlinear RG3 injections (compensated pair-writes with arbitrary conjunction fragments).